

Evasive Reverse Shell

Windows | ERS-W | Stealth Assessment Report

DISCLAIMER: This tool is intended for authorized security testing, CTF challenges, and educational purposes only. Unauthorized use is illegal.

Stealth Rating
24/24

Platform	Windows
Binary Size	2.2 MB
Post-Build Patches	494
Entropy	6.82
XOR Key	0xAB
Language	Rust

WSS/TLS reverse shell for Windows with sandbox detection, 18 builtins, registry persistence, and sleep mask encryption.

Stealth Audit Results

#	Check	Result
1	Tool/command strings	PASS
2	Privilege names	PASS
3	Registry/CVE strings	PASS
4	Output messages	PASS
5	Build/debug paths	PASS
6	Library strings (rustls/TLS/OpenSSL)	PASS
7	Entropy (6.82)	PASS
8	Anti-debug (IsDebuggerPresent)	PASS
9	Self-delete (batch script)	PASS
10	Binary size (2.2 MB)	PASS
11	No packer signatures	PASS
12	Stripped symbols	PASS
13	Process masquerade	PASS
14	Network/C2 indicators	PASS
15	YARA rule triggers	PASS
16	File metadata	PASS
17	Suspicious API imports	PASS
18	Runtime string leaks	PASS
19	XOR string encoding	PASS

20	Compilation flags	PASS
21	User-Agent encoding	PASS
22	Sandbox detection	PASS
23	Analysis tool names	PASS
24	Persistence/credential strings	PASS

Defense Evasion Techniques

Technique	Description
XOR String Obfuscation	All sensitive strings encoded at compile time (key=0xAB)
Binary Patching	494 post-build patches scrub rustls/TLS/OpenSSL/build paths
Sandbox Detection	6-check weighted scoring: process count, uptime, tools, disk, RAM, username
Anti-Debug	IsDebuggerPresent() - exits silently if debugger attached
Process Masquerade	Console title: Windows Update Host
Self-Deletion	Spawns delayed batch script to remove binary
Sleep Mask Encryption	XOR-encrypt memory markers during sleep intervals
Minimal Binary	opt-level=z, LTO, stripped, panic=abort

Command & Control

Feature	Description
WSS over TLS	WebSocket Secure with rustls (pure Rust TLS)
Chrome User-Agent	Chrome 120 header impersonation (XOR encoded)
Exponential Backoff	5s initial to 5min max with 30% jitter
Auto-Reconnect	Infinite reconnection loop with backoff

Built-in Commands

Command	Description
whoami	Current user (env-based)
hostname	Computer name
pwd / cd	Working directory
ps	Process list
kill [pid]	Kill process
info	System info beacon
persist	Registry Run key
pe	Inline privesc scan
sleep [sec]	Encrypted sleep
exit	Self-delete + exit

MITRE ATT&CK; Mapping

ID	Technique
T1071.001	Web Protocols
T1573.002	Encrypted Channel
T1027	Obfuscation

T1036.004	Masquerading
T1070.004	Indicator Removal
T1622	Debugger Evasion
T1497	Virtualization/Sandbox Evasion
T1059.003	Command Interpreter (cmd.exe)
T1547.001	Registry Run Keys
T1082	System Information Discovery
T1057	Process Discovery

Project Files

File	Purpose
src/main.rs	Implant source code
build.sh	Build + binary scrub script
scrub.py	494-patch same-length binary patcher
listener.py	WSS C2 listener (same as ERS)
win_rev_shell_wss.exe	Pre-built scrubbed binary (2.2 MB)

Companion Projects

Project	Platform	Stealth
ERS	Linux	20/20
ERS-W (this)	Windows	24/24
SPES	Linux	20/20
SPES-W	Windows	24/24

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